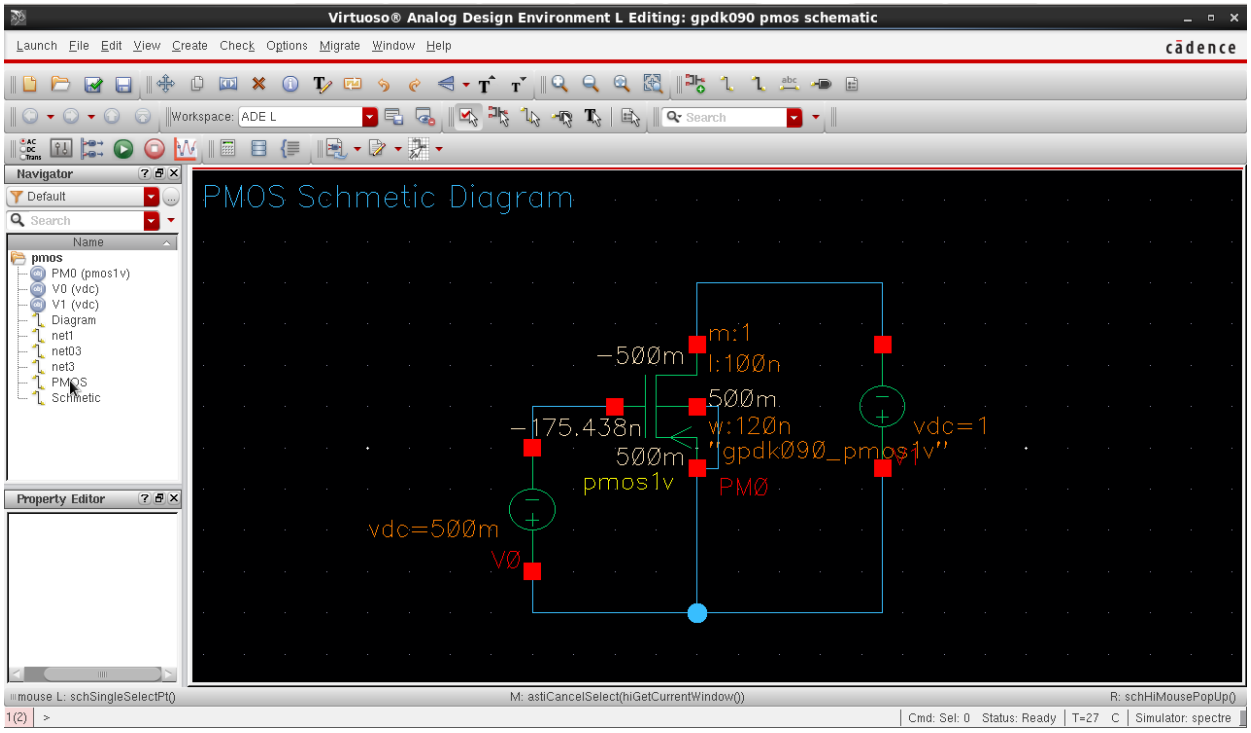
Assignment - 01

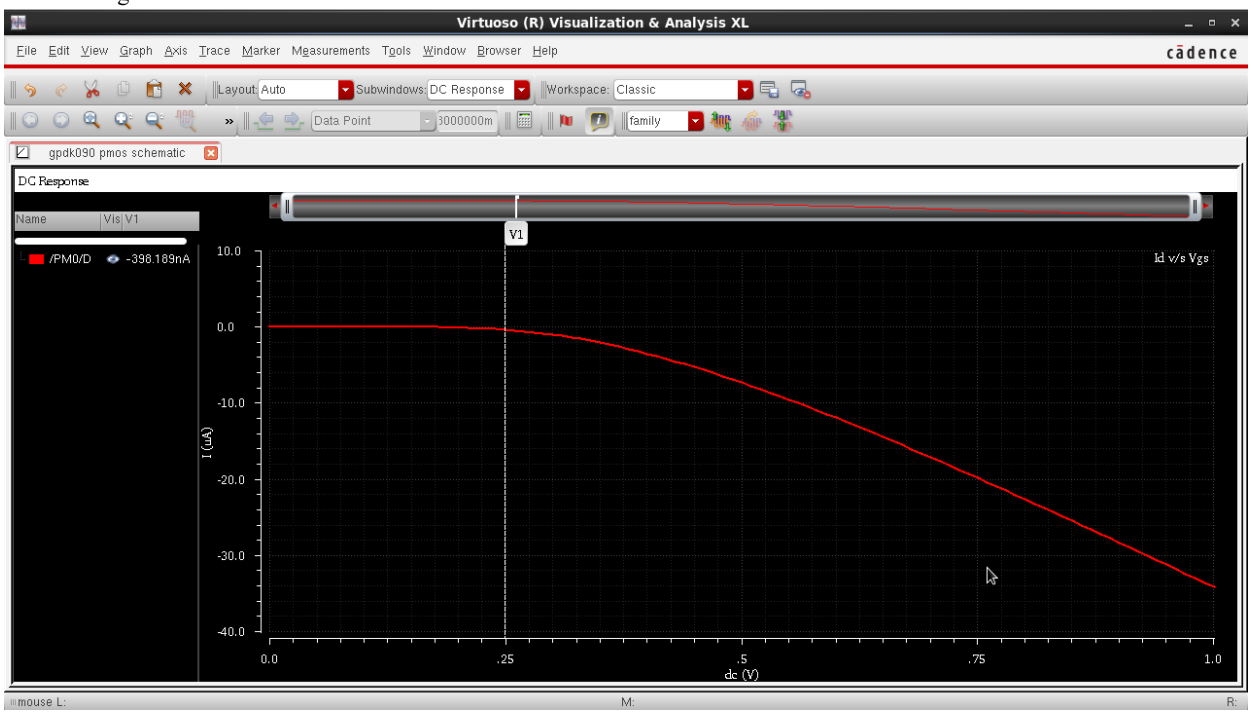
DC analysis of PMOS

1. Schematic diagram of PMOS:



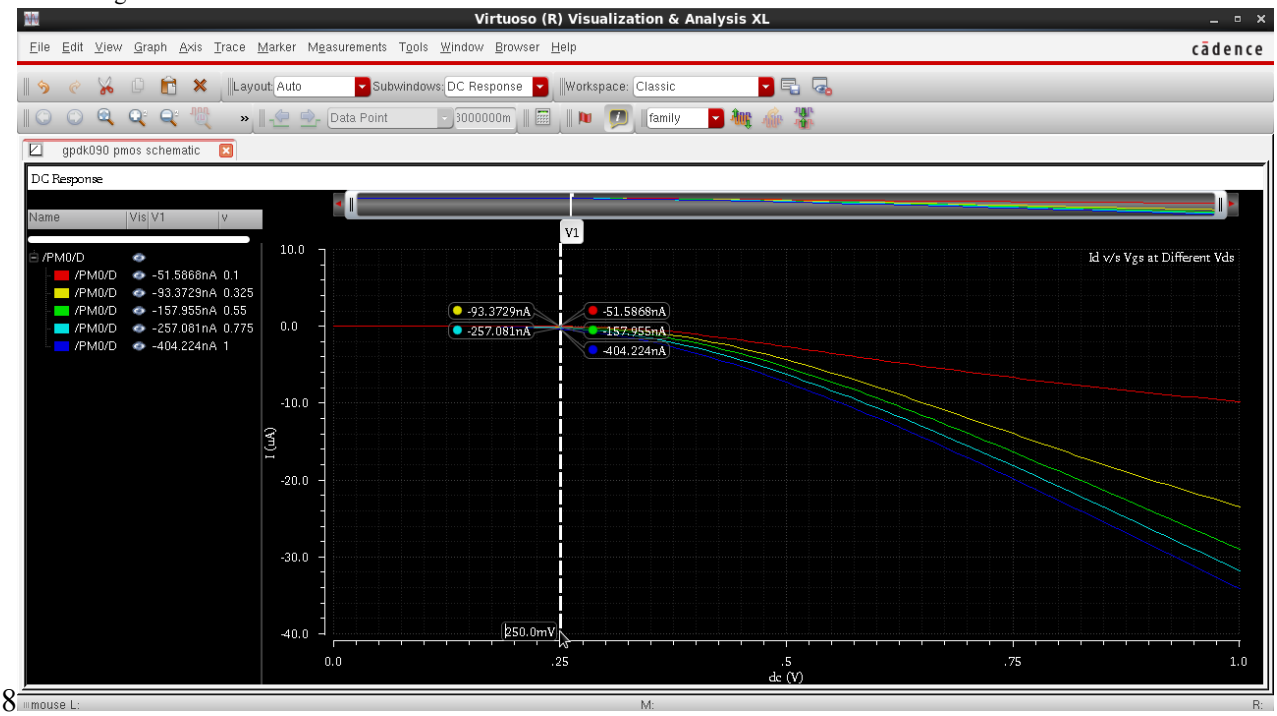
2. I-V Characteristics:

- I_d v/s V_{gs}:



- Initially, the PMOS is OFF because V_{gs} is below the threshold, so the drain current is nearly zero.
- As V_{gs} exceeds the threshold, a channel forms and the drain current increases in magnitude.

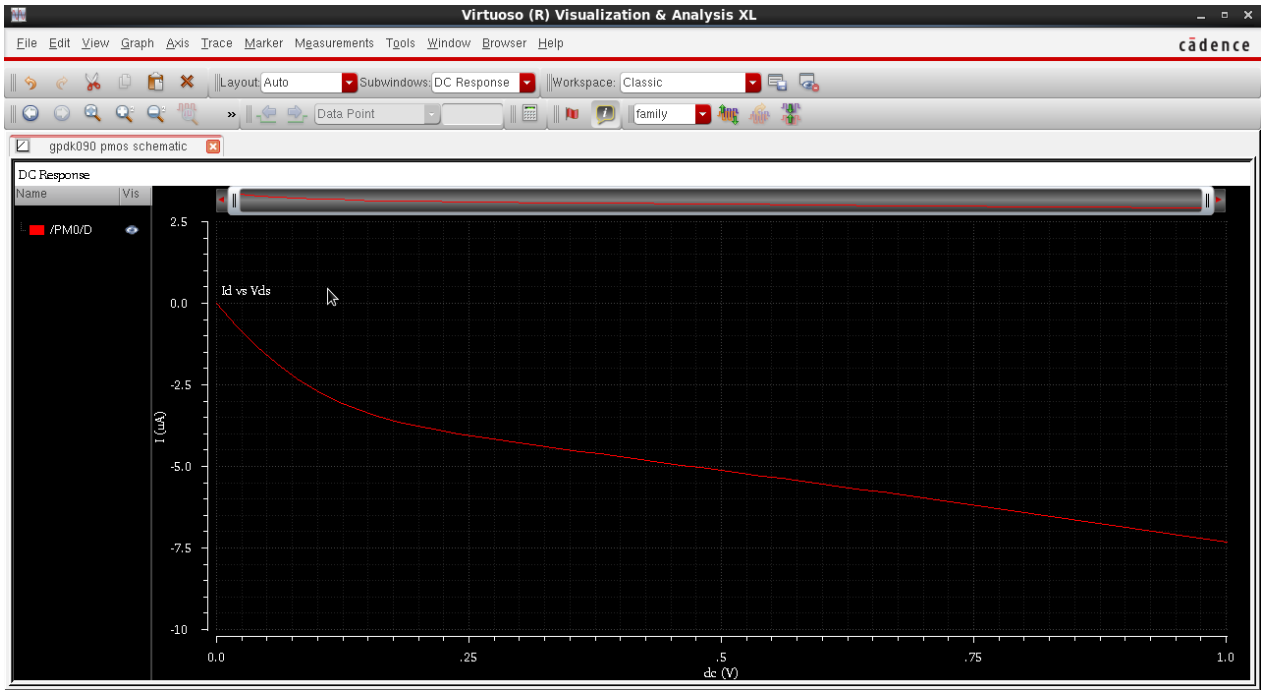
- I_d v/s V_{gs} w. r. t. different of V_{ds}:



8

  - As V_{ds} increases, the magnitude of the drain current increases for the same V_{gs}, so the curves shift downward."
  - This is because a larger drain-to-source voltage creates a stronger electric field, allowing more hole current to flow through the PMOS channel.

- I_d v/s V_{ds}:



  - The drain current (I_d) increases with increasing drain-to-source voltage (V_{ds}) while the gate voltage is kept constant."
  - At higher V_{ds}, the rate of increase of I_d becomes smaller as the NMOS transistor approaches the saturation region.

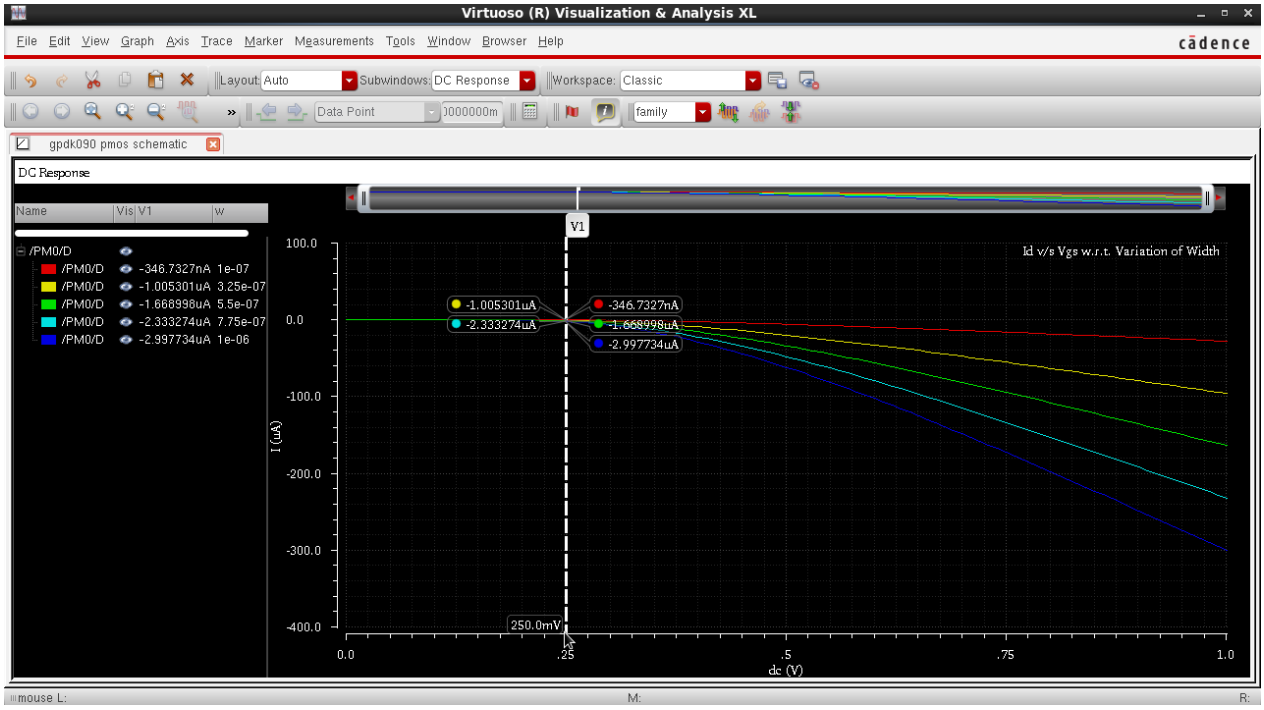
- I_d v/s V_{ds} w. r. t. different of V_{gs}:



  - As V_{DS}increases, the drain current initially increases rapidly and then enters the saturation region, where the current changes only slightly.
  - Increasing V_{gs}forms a stronger inversion channel, so the magnitude of the drain current ($|I_d|$) increases for the same V_{ds}.

3. I-V Characteristics w. r. t. Variation of Parameter:

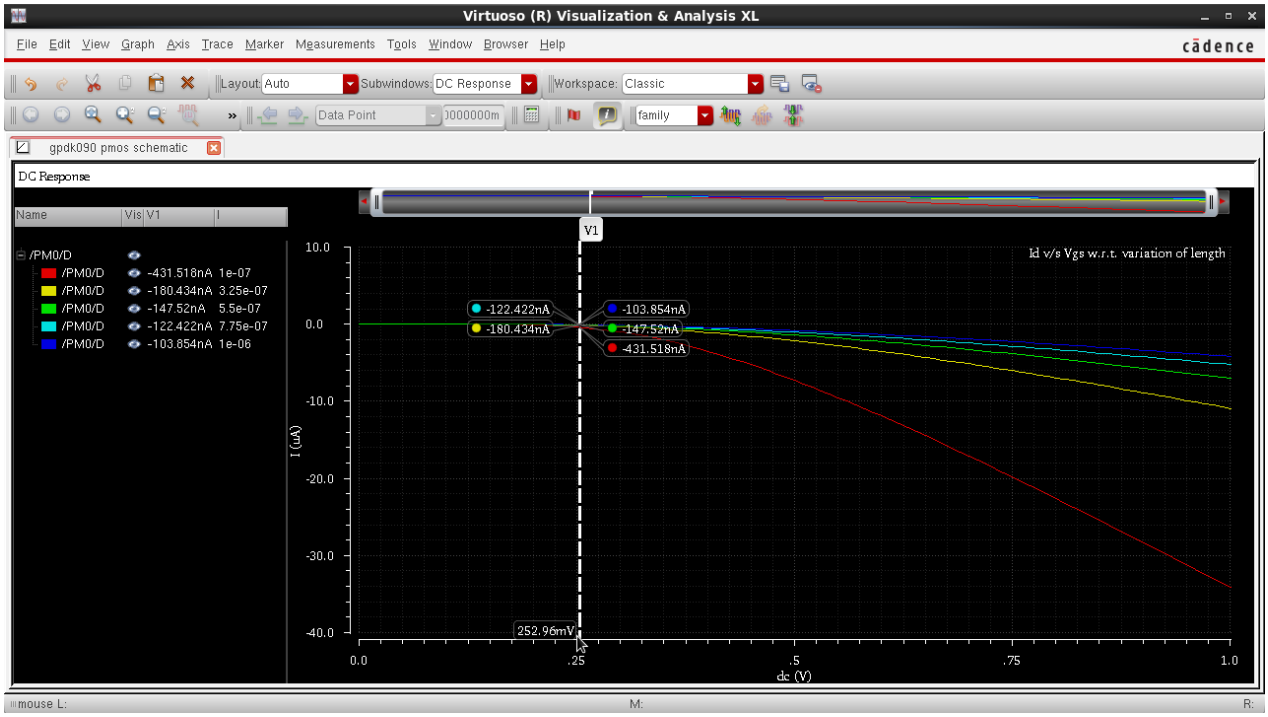
- I_d-V_{gs} Characteristics w. r. t. Variation of Width:



  - As the channel width (W) increases, the drain current increases because a wider channel provides more conduction paths for charge carriers.

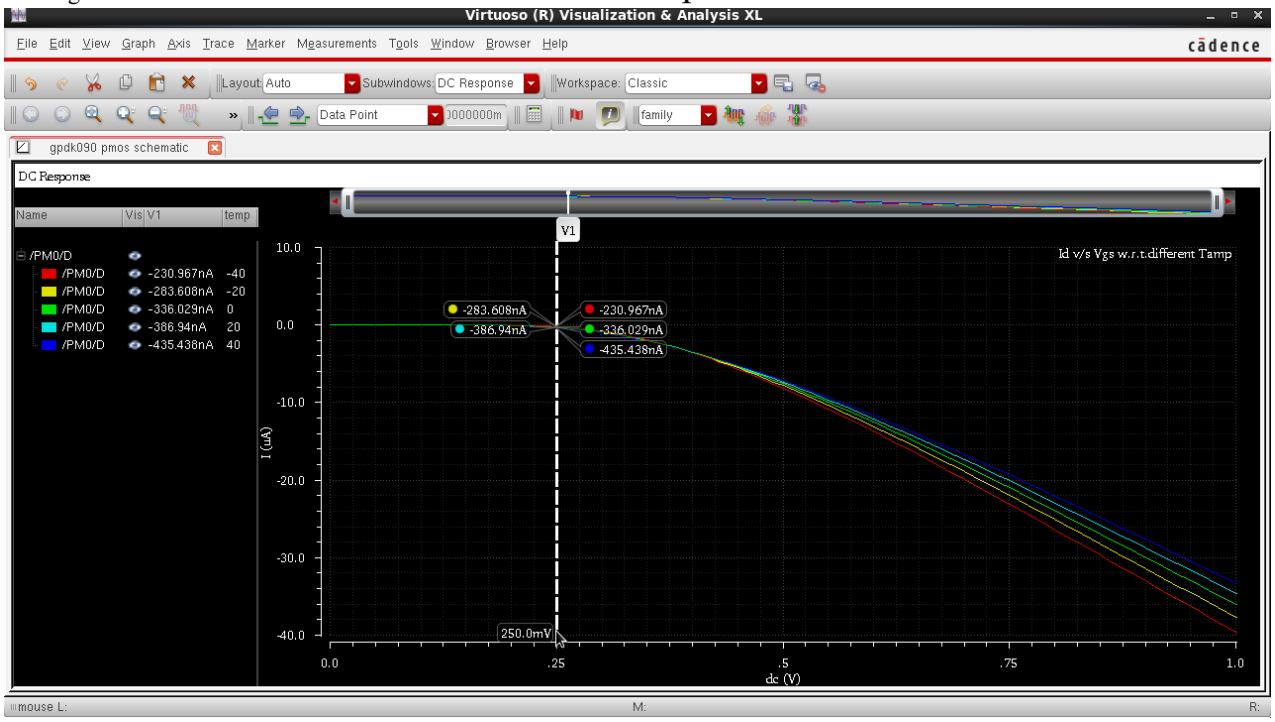
- Therefore, for the same V_{gs}, devices with larger width produce a higher magnitude of drain current ($|I_d|$).

- I_d-V_{gs} Characteristics w. r. t. Variation of Length:



- As the channel length (L) increases, the drain current decreases because the channel resistance increases.
- Therefore, for the same V_{gs}, devices with shorter channel length produce a higher magnitude of drain current."

- I_d-V_{gs} Characteristics w. r. t. Variation of Temperature:



- As the temperature increases, the drain current decreases because carrier mobility reduces due to increased lattice scattering, which lowers the channel conductivity.